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# AEFETA: Encrypted traffic classification framework based on self-learning of feature

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**Abstract—This paper proposes an end-to-end framework based on feature self-learning, AEFETA, to handle the task of encrypted network traffic classification. In the AEFETA, we propose a data preprocessing method based on the structure information of the network traffic, and at the same time, a lightweight deep learning model combined with an attention mechanism also be proposed in this framework. We can convert the raw pcap file into a file with the data type that can be input to the deep neural network through data preprocessing, and then, automatically extract the spatial-temporal features perform classification tasks. AEFETA framework achieved a recall of 0.98 and precision of 0.97 in the encrypted traffic classification task, meanwhile, we find the best data format through experimental comparison.**

***Keyword-deep learning; attention; end-to-end framework***

## I. Introduction

Due to the increasing awareness of user privacy protection and the scalability and ease of operation of traffic encryption technology, encrypted traffic accounts for an increasing proportion in the real world. Whether to use a secure encryption protocol has become an important indicator for evaluating a website. Google uses HTTPS as a reference factor for search engine ranking. According to Netmarketshare data[1], as of October 2019, the proportion of global encrypted traffic has exceeded 90%. How to accurately identify encrypted traffic has become the hardest challenge of our time. The current traffic classification technology can be divided into the following three categories:

- Fingerprint-based approach. Traffic classification through port number and DPI(deep packet inspection) is collectively referred to as a fingerprint-based method. The port-based classifier is too old and is no longer used. The DPI-based classifier is currently the main method in the industry, which analyzes the payload fields for traffic classification. However, it cannot identify unknown categories. In the encrypted traffic classification task, it can only obtain information from unencrypted data such as the protocol packet header, which is difficult to complete the classification task.
- Statistics-based approach. This method usually uses flow statistics, packet statistics, or behavior statistics to classify. After feature extraction, machine learning algorithms such as support vector machines [2] and random forests [3] are used to train and test feature vectors. Statistics-based methods can effectively identify unknown categories, but how to extract effective features is an issue worthy of research. At the same time, fixed features are easy to be attacked is also an ongoing problem.
- Deep-learning-based approach [4]. The method based on deep learning is a research hotspot that has appeared in the field of traffic classification in recent years. [5] proposes to use CNN to learn the feature of network traffic to achieve end-to-end traffic classification. This way can avoid a lot of feature engineering, but, because deep learning is a data-driven method, the collection of a dataset in each sub-field of traffic classification is difficult. data imbalances in the existing dataset influence classification results.

At present, deep learning technology has a wide range of applications in the field of text recognition. It can abstract and extract the features of samples layer by layer through the multi-layer neural network structure and the adjustment of a large number of parameters. Network traffic data has hierarchical features similar to text (network traffic data: sessions, packets, bytes; text data: paragraphs, sentences, words). Therefore, To solve the problem of difficulty in feature selection and fixed features easily targeted by attackers, this article focuses on the common method of text classification, deep learning, to research encrypted network traffic classification task. The main contributions of this article are as follows:

- A new classification framework AEFETA is proposed. Use deep learning technology to automatically learn the temporal and spatial features of network traffic, and identify encrypted network traffic from a global perspective. The classification framework does not rely on any prior knowledge about the protocol and topology, nor does it require manual feature selection.
- an attention-based spatiotemporal feature extraction algorithm is proposed. Given the different degree of importance distribution of each session and each packet information, the algorithm uses the attention mechanism [6] to weight the extracted features according to the importance, thereby improving the classification accuracy.

- A data processing method is proposed. We use experimental comparison to find the data representation with the best classification effect.

## II. Related work

Encrypted traffic refers to the traffic generated by the encryption algorithm transmitted during the communication process. [13-15] conducted related research on encryption and unencrypted, and proposed various methods for traffic identification, which achieved good results. However, as the demand for traffic analysis increases, only identifying whether the traffic is encrypted cannot meet the requirements. The current research on encrypted traffic classification is mainly divided into two categories: anomaly identification and application service identification. Among them, abnormal identification is to identify malicious traffic such as DDoS and Botnet. Application service identification is to identify the type of service to which encrypted traffic belongs, such as Browning, Streaming, Chat, etc.

### *A. anomaly identification*

Although data encryption can effectively deal with man-in-the-middle attacks to protect data, it can also pose new security threats and risks. Cyren[7] found that 37% of malware uses HTTPS, and every major ransomware family has spread through HTTPS. Therefore, it is urgent to detect encrypted malicious traffic. [8-10] proposed a variety of methods to identify malware traffic using machine learning methods. For the TLS protocol, Cisco [11-12] used logistic regression and ten-fold cross-validation to identify encrypted malicious traffic, with an accuracy of over 90%. Ivan et al. [16] analyzed and summarized 40 data characteristics in 4 categories, and used LSTM and 5-fold cross-validation to detect malware and phishing software signatures. Frantisek et al. [17] compared the four machine learning methods of XGBoost, Random Forest, Neural Network, and SVM for the detection of TLS encrypted malicious traffic, and concluded that the XGBoost and Random Forest methods are more effective. Zeng et al. [22] proposed a DFR model, which slices the original data according to 900 bytes and converts it into a 30×30 two-dimensional matrix, and then uses the three-layer structure of CNN, LSTM, and SAE for traffic identification.

### *B. application service identification*

Certain network management tasks require fine-grained identification of service types for encrypted traffic. For example, the use of social networking sites is not allowed in certain exams, but applications such as web pages and videos are not restricted. [18] compared three P2P traffic identification methods, including identification methods based on ports, application layer signatures, and flow statistics. [20] proposed a method called Skype. Hunter's method recognizes Skype traffic in real- time. This method can effectively identify signaling services and data services (such as voice, video, and file transmission) through a strategy based on a combination of signature and stream statistical characteristics. [21] Divide the conversation into 784-byte slices, convert them into grayscale images, and enter CNN for classification. Experiments show that the effect of using 1D-CNN is better than 2D-CNN. [23] proposed a deep packet framework combining 1D-CNN and SAE. [19] proposed a bidirectional LSTM model (HABBiLSTM) based on a hierarchical attention mechanism to introduce attention to the traffic classification task for the first time, and achieved good classification results.

Although the deep learning method based on feature self-learning can realize effective end-to-end traffic classification, data processing and model construction still need more research.

## III. Methodology

In this work, we developed a framework, AEFETA, that comprises two parts, data processing, and a deep learning model.

### *A. data processing*

This paper uses the original pcap format file as input data and completes the data preprocessing through the process of Figure 1 to obtain a set of data, which directly input to the model.

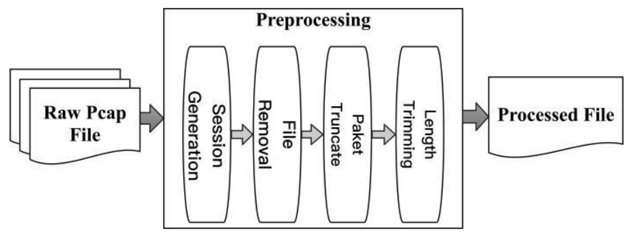


Fig.1. The process of data preprocessing

Step 1: Session generation. Session generation refers to splitting a piece of the original file into multiple files, each of which is a session. We use the segmentation tool in the USTC-TK2016 toolset to complete this step.

Step 2: File removal. This step removes the empty files and UDP files.

Step 3: Paket truncate. In this step we remove 16 bytes packet header, 34 bytes link layer and network layer information, and, the 4 bytes port number information in the transport layer. We think this information is not important or even disturbing. The120 bytes after these 54 bytes as useful data.

Step 4: Paket trimming. This step saves the first 4 packets with the application layer。

### *B. deep learning model*

Our deep learning model for encrypted traffic classification is composed of CNNs and BiLSTM with attention mechanism. Since the location of the data packet header information is fixed, we think it has certain spatial characteristics; the process of key exchange in the early stage of the data stream also maps the behavior pattern of a data stream. Therefore, this article automatically extraction of spatio-temporal characteristics of network traffic data to classify traffic. Since each data packet in the data stream is of different importance, we introduce the attention mechanism into our deep learning model (Figure 2).

The algorithm1 is describe the architecture of deep learning model in the AEFETA. This process is mainly divided into four layers: spatial feature learning layer, temporal feature learning layer, attention layer, and output layer. It should be noted that the model input is an n*m size matrix, and the output is the predicted category.

*1) Spatial feature learning layer*

Since different convolution kernels will learn different

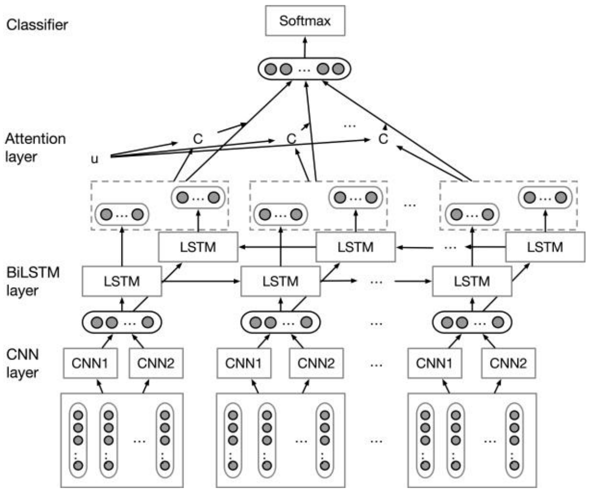


Fig.2.deep learning model

**Algorithm 1** AEFETA

**Input:** preprocessed data streaming $S = \{p_i | i = 0, 1, ..., n\}$
**Output:** predicted label

**for** each $p_i \in S$ **do**
  $\overrightarrow{h_{i1}} = \overrightarrow{cnn_1}(p_i)$  ▷ $p_i = \{d_j | i = 0, 1, ..., m\}$
  $\overrightarrow{h_{i2}} = \overrightarrow{cnn_2}(p_i)$
  $\mathbf{h}_i = [\overrightarrow{h_{i1}}, \overrightarrow{h_{i2}}]$
  $\overrightarrow{v_i} = \overrightarrow{lstm}(\mathrm{h}_i)$
  $\overleftarrow{v_i} = \overleftarrow{lstm}(\mathrm{h}_i)$
  $\mathbf{v}_i = [\overrightarrow{v_i}, \overleftarrow{v_i}]$
**end for**
$\mathbf{c} = \sum_{i=1}^{n} (\alpha_i v_i)$
**predict**=softmax($[v_1, ..., v_n, c]$)

spatial features, applications in the field of computer vision have shown that CNN using different convolution kernels connected will get better spatial features. In the spatial feature learning layer of the model, we use two CNN with different convolution kernel sizes to complete spatial feature learning. The specific steps are as follows:

Convert each data packet into one-hot encoding form, and convert the one-dimensional array into a matrix $\mathbf{p_i}$ of m*256 sizes.

- Input $\mathbf{p_i}$ into two convolutional neural networks with different convolution filters to learn the spatial characteristics of the data packet. The output vectors of the two neural networks are $\mathbf{h_{i1}}$ and $\mathbf{h_{i2}}$ respectively.
- splicing $\mathbf{h_{i1}}$ and $\mathbf{h_{i2}}$ to obtain the packet vector $\mathbf{h_i}$.

*2) Temporal feature learning layer*

BiLSTM can learn contextual information better and has been widely used in tasks such as text classification. In the temporal feature learning layer of the model, we use BiLSTM to complete the learning of temporal features. Specific steps are as follows:

The vector sequence $[\mathbf{h_1}, \mathbf{h_2}, \ldots, \mathbf{h_n}]$ of the data packet is sequentially inputted into the BiLSTM to learn the temporal features to obtain the vector $\overrightarrow{v_i}$.

The data packet vector sequence is input in the reverse order $[\mathbf{h_n}, \mathbf{h_{n-1}}, \ldots, \mathbf{h_1}]$ into the learning temporal feature of the LSTM to obtain the vector $\overleftarrow{v_i}$.

- Splicing $\overrightarrow{v_i}$ and $\overleftarrow{v_i}$ get the output vector $\mathbf{v_i}$.

*3) Attention layer*

The attention mechanism is widely used in deep learning tasks with Encoder-Decoder structure such as image processing and speech recognition. Its core goal is to find out more critical information for the current task from a large amount of information. In the attention layer of the model, we perform a weighted summation on the output vector $\mathbf{v_i}$ of the upper layer to obtain the feature vector $\mathbf{c}$:

$$c = \sum_{i=1}^{n} (\alpha_i v_i) \tag{1}$$

Among them, the calculation formula of αi is:

$$\alpha_i = \frac{exp(u_i u)}{\sum_i exp(u_i u)} \tag{2}$$

***u*** is the random initialization vector for subsequent training, and the formula for calculating $\boldsymbol{u_i}$ is:

$$u_i = tanh(W v_i + b) \tag{3}$$

*4) Output layer*

In the field of deep learning, softmax is a commonly used classification function, especially in multiclassification scenarios. Softmax maps some inputs to functions between 0-1 and normalizes the final prediction vector to ensure that the sum is 1. In the output layer of the model, we use softmax as the classifier of the model. Since encrypted traffic recognition is a multi-classification problem with the imbalance of categories, we choose cross-entropy as the loss function of the model.

## IV. Experiments and Discussion

The hardware and software configuration of the experiment is as follows. The software framework is Keras and TensorFlow, running on the Ubuntu operating system. In terms of experimental hardware, a 64-core 256G cpu is used. Besides, an Nvidia GeForce RTX 2080Ti gpu is used as a training accelerator. Table1 describes the architectures of the deep learning model.

TABLE1. NEURAL NETWORK ARCHITECTURAL PARAMETERS

| Layer | Type | Filters/Neurons/Rate |
|---|---|---|
| 1 | Conv+tanh | 128 |
| 2 | dropout | 0.2 |
| 3 | GlobalMaxPool | / |
| 4 | dense | 128 |
| 5 | Conv+tanh | 256 |
| 6 | dropout | 0.2 |
| 7 | GlobalMaxPool | / |
| 8 | dense | 128 |
| 9 | concatenate | / |
| 10 | dense | 64 |
| 11 | Bilstm | 64 |
| 12 | attention | / |
| 13 | dense | 6 |
| 14 | softmax | / |

### A. Dataset

This article uses the public data set VPN-nonVPN data set (ISCXVPN2016) [6] to carry out the encrypted traffic classification experiment proposed in this article. The data set includes 14 kinds of encrypted traffic (7 kinds of protocol encapsulated traffic and 7 kinds of conventional encrypted traffic). This experiment only discusses conventional encrypted traffic data. Because of the confusion in the classification of some of the files, for example, facebook_video and other files can be classified as browser or streaming, so we ignore these files and divide 61 files into 6 categories, namely: chat, email, file_transfer, p2p, streaming, voip(Table2 ).

TABLE2. ISCXVPN2016

| label | File |
|---|---|
| chat | AIMchat1, aim_chat, facebookchat, hangouts_chat, ICQchat, icq_chat, skype_chat |
| email | email |
| file_transfer | skype_file |
| p2p | Torrent01 |
| streaming | netflix, spotify, vimeo, youtube |
| voip | hangouts_audio, voipbuster |

### B. Evaluation results

In this section, we discuss the results of the AEFETA framework. Generally speaking, detection accuracy, accuracy rate, recall rate, and F1 value are the four key indicators for evaluating encrypted traffic recognition. This article uses the following criteria to evaluate the classification algorithm: precision, recall, F1-Score.

$$Ac(Accuracy) = \frac{TP+TN}{TP+FP+FN+TN}$$
$$Pr(Precision) = \frac{TP}{TP+FP}$$
$$Re(Recall) = \frac{TP}{TP+FN}$$
$$F1(F1-score) = \frac{2PR}{P+R}$$

Our proposed framework performs well than other encrypted traffic classification methods in the same dataset. Table3 shows the comparision results of the[21], [23], and ours. Figure 3 shows the precision, recall, F1-Score value of each class of traffic, all three indicators of the streaming label reach 1.

TABLE3. COMPARISON OF DIFFERENT METHODS

| **Methodology** | Ac | Pr | Re | F1 |
|---|---|---|---|---|
| [21] | 0.817 | 0.855 | 0.858 | 0.856 |
| [23] | / | 0.97 | 0.97 | 0.97 |
| ours | 0.984 | 0.979 | 0.980 | 0.979 |

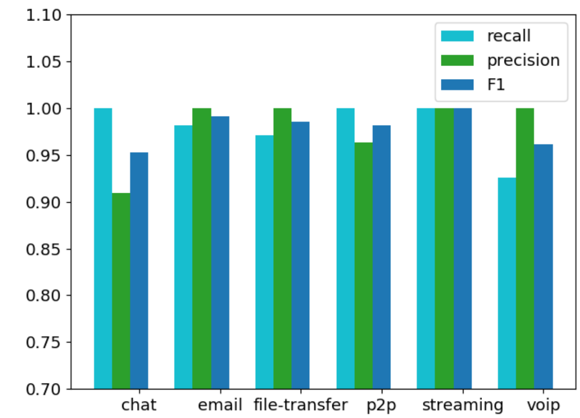


Fig.3. Recall, precision, F1 value of 6-class classifiers

### C. discussion

In this section, we discuss the impact of the number of packets in a session and the length of the packets on the classification results. Because the larger input format will greatly reduce the speed of the algorithm, we set the number of packets in the range of 4-6, and the packet length in the range of 40-140 for experimentation. We compared the recall value and accuracy value under different packet lengths and packet numbers. Through the display of Figure 4 and Figure 5, we choose 4 and 120 as the number of packets per session and length of per packet.

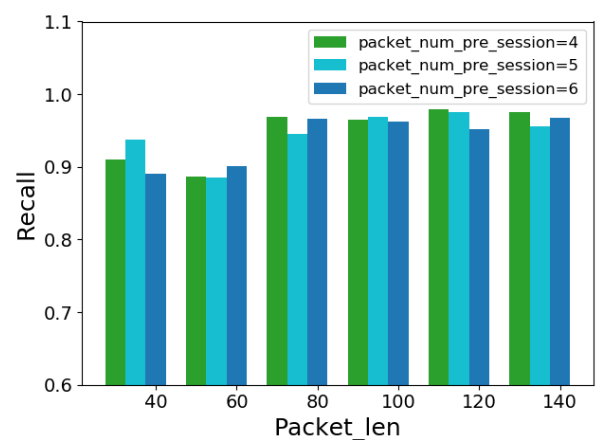


Fig.4. The recall value under different packet lengths and packet numbers

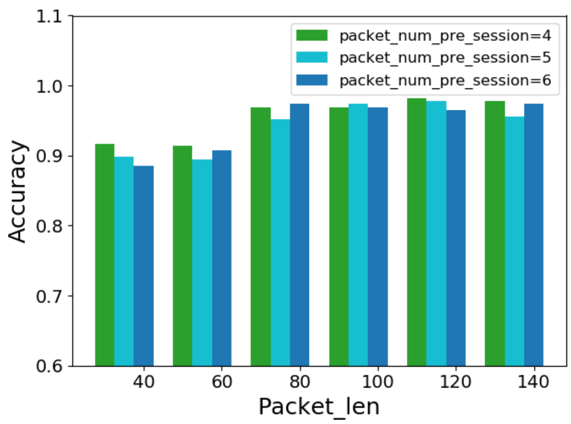


Fig.5. The accuracy value under different packet lengths and packet numbers

## V. Conclusion

In this paper, we presented a framework, AEFETA, that automatically extracts features from encrypted traffic for traffic classification tasks. This paper uses the hierarchical characteristics of network traffic, draws on the application of attention mechanism in text classification, and uses deep learning to directly process the original pcap file to achieve end-to-end traffic classification. Our experiment was carried out on vpn-nonvpn and achieved good results. At the same time, we discussed data processing, which has a particularly great impact on classification results. What this article is currently doing is classifying ordinary encrypted traffic. In the field of malware detection, how to perform malware detection on encrypted traffic while protecting user privacy (that is, without decrypting data packets) is a difficult point, it is also a problem that needs to be research in the next step.

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